

Do Running Gait Biomechanics Differ Between Adults and Adolescents with Patellofemoral Pain?

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INTRODUCTION

Patellofemoral pain (PFP) is defined by pain around the kneecap that is aggravated during activities on a flexed knee [1]. During running, adults with PFP have slower velocity and greater contralateral pelvic drop compared to adults without PFP [1]. In contrast, adolescents with PFP run with a higher cadence and a greater knee flexion angle at 35-40% of the gait cycle compared to adolescents without PFP [2].

Adolescents with PFP often experience poorer treatment outcomes from strength training compared to adults with PFP [3]. Therefore, aimed to compare biomechanical data of adults and adolescents with and without PFP during running.

METHODS

Biomechanical data were collected during running in 10 adolescents and 10 adults with PFP and compared to 10 adolescents and 10 adults without PFP. Participants with PFP had their diagnosis confirmed by a registered allied health practitioner. Participants completed two 4-minute running trials on a dual-belt treadmill (Bertec, Ohio, USA). One running trial was completed in a conventional shoe (Asics Gel Cumulus, Asics, Kobe, Japan) and another in a flat flexible shoe (Vivobarefoot RA II, Vivobarefoot, Freiburg, Germany). The order of conditions were counterbalanced. All participants self-selected a comfortable running speed.

A 10-camera VICON MX motion analysis system (Oxford Metrics, Oxford, UK) sampling at 250 Hz was used to capture three-dimensional (3D) joint kinematics of the lower limbs. GRF data were collected simultaneously, sampling at 1000 Hz. Kinematic data were collected using 32 retroreflective markers in accordance with a previously established lower-limb model developed by McLean et al. [3].

Twenty gait cycles were extracted from the most affected limb or dominant limb, using a 30-second period in the fourth minute of each trial. Joint kinematics and kinetics were computed using 3D marker coordinates and force plate data using inverse kinematics and dynamics tools in OpenSim 4.5 (OpenSim, California, USA). Joint moments were normalised to body mass and reported in Nm/kg. Data was processed and analysed using custom written Python 3.12 (Python Software Foundation, www.python.org) scripts.

RESULTS AND DISCUSSION

Results are currently being analysed and will be presented in full at the SENS Symposium 2026. Descriptive statistics are presented in Table 1. Adults were significantly older and ran significantly faster compared to the adolescent cohorts (Table 1). No other variables were significantly different.

We hypothesised that there would be biomechanical differences between adults and adolescents with PFP which may cause the varied responses to recommended treatment options.

CONCLUSIONS

These findings will provide insight into whether gait biomechanics are similar between adults and adolescents both with and without PFP. This may inform whether it is appropriate to provide the same treatment to both PFP cohorts..

REFERENCES

- [1]Bazett-Jones, D et al. *Sport Med* **53** 519-547, 2023
- [2]Mazella, N et al. *Scand J Med Sci Sport* **34** 1-8, 2024
- [3]van Middelkoop, M *JOSPT* **47** 801-805, 2017
- [4]McLean, S et al. *J Biomech Eng* **125** 864-874, 2003

Table 1: Descriptive statistics for adults and adolescent cohorts both with and without PFP.

	Adolescent PFP	Adolescent Control	Adult PFP	Adult Control	p-value
Age, years	14.32 (1.63)	14.08 (1.61)	26.5 (6.11)	27.04 (6.59)	<0.001
Sex, male (%)	16 (57.14)	13 (54.17)	10 (38.46)	10 (38.46)	0.375
Height, cm	167.85 (9.96)	170.85 (9.66)	169.48 (8.56)	168.67 (8.54)	0.687
Body mass, kg	64.01 (9.96)	63.91 (13.57)	68.81 (12.65)	67.17 (12.16)	0.429
Duration of symptoms, months	24.7 (20.2)	N/A	27.66 (37.70)	N/A	0.662
Worst Pain	6.5 (1.7)	N/A	5.96(1.40)	N/A	0.156
Usual Pain	3.6 (1.5)	N/A	3.38(1.18)	N/A	0.621
Running speed	2.39 (0.20)	2.43 (0.24)	2.66 (0.23)	2.60 (0.23)	<0.001

Note: values are presented as mean (SD) unless stated otherwise. Worst and Usual pain are measured on an 11-pt numerical rating; 0=no pain, 10=worst pain.